

A Lightweight Custom CNN for Multi-Class Classification of Kidney Abnormalities from

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Abstract—Medical image classification is a critical task that assists radiologists in rapid and accurate diagnosis. While large pre-trained Convolutional Neural Networks (CNNs) are popular, they are often computationally expensive and prone to overfitting on specific grayscale medical datasets. This paper presents a comparative study of standard architectures (VGG16, ResNet-50, EfficientNet) and proposes a custom, lightweight CNN optimized for classifying kidney CT scans into four categories: Normal, Cyst, Tumor, and Stone. Utilizing a dataset of 12,446 verified CT images, the proposed model achieved an overall accuracy of 95.50% and an inference time of 5.45 seconds, significantly outperforming heavy baseline models in both computational efficiency and diagnostic reliability.

Keywords—Convolutional Neural Networks, CT Scans, Deep Learning, Medical Imaging, Kidney Abnormalities.

I. INTRODUCTION

The global scarcity of specialized nephrologists and radiologists has accelerated the need for AI-driven diagnostic systems. Computed Tomography (CT) is a primary modality for detecting renal abnormalities, but manual interpretation is time-consuming and subject to human error. Developing automated, end-to-end deep learning models can standardize kidney monitoring and reduce the burden on clinicians. However, standard deep learning architectures require immense computational resources. This project investigates the efficacy of existing CNN architectures and proposes a lightweight custom CNN to accurately classify multi-class kidney abnormalities while minimizing computational overhead.

II. RELATED WORK

Recent literature has explored various deep learning models, ranging from traditional CNNs to Vision Transformers, for classifying renal diseases. A summary of highly relevant studies on the same or similar datasets is presented in Table I.

TABLE I. SUMMARY OF RELATED WORK

Ref	Model	Accuracy	Pro	Cons
[1]	YOLOv8n-cls	82.52%	Uses a single CNN, saving CPU resources; standardizes monitoring.	High dependence on annotation quality; recall of 75.28% indicates margin of error.
[2]	Vision Transformers (Swin, CCT, EANet) & CNNs (VGG16, ResNet50). Used GradCAM.	Swin: 99.30% VGG16: 98.20% CCT: 96.54%	Swin Transformer was highly accurate; GradCAM proved models focused on correct ROI.	Severe data balancing required (limited to 1,300 images/class); ResNet50

		EfficientNet: 77.02% ResNet50: 73.80%		performed poorly.
[3]	Marker-based Watershed Segmentation + Customized 16-layer CNN	Customized CNN: 98.66% ResNet50: 87.92% EANet: 83.65%	Highest sensitivity/specificity; Watershed algorithm effectively isolates the ROI.	Watershed is sensitive to noise, causes over-segmentation, and struggles with low-contrast boundaries.
[4]	End-to-end lightweight CNN (Stacked Conv + GAP)	99.80%	Near-perfect accuracy across diverse datasets; highly reliable for portable/home setups.	Requires roughly twice the calculations of a basic CNN to generate effective feature maps.

III. DATASET DESCRIPTION

The data utilized in this study is the "CT KIDNEY DATASET: Normal-Cyst-Tumor and Stone". It was collected from Picture Archiving and Communication Systems (PACS) across different hospitals in Dhaka, Bangladesh, containing diagnosed cases of kidney tumors, cysts, normal, or stone findings. Each image finding was verified by both a radiologist and a medical technologist to ensure ground-truth correctness. The finalized dataset contains 12,446 unique images distributed as follows:

- Normal: 5,077 images
- Cyst: 3,709 images
- Tumor: 2,283 images
- Stone: 1,377 images

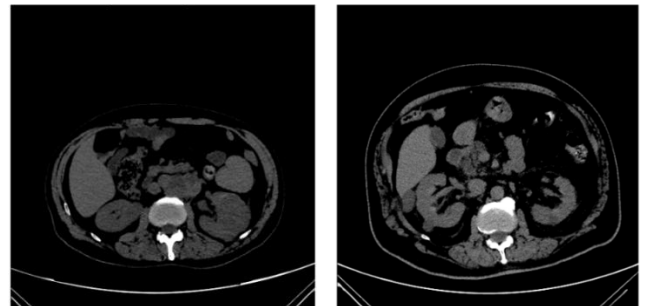


Fig. 1. Sample CT Images of Cyst

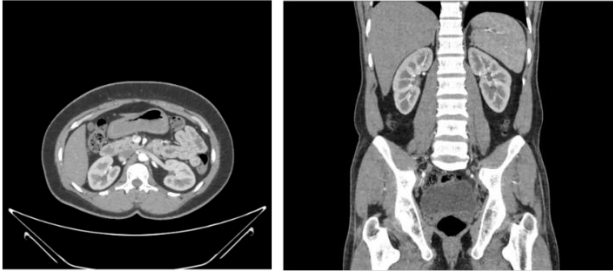


Fig. 2. Sample CT Images of Normal

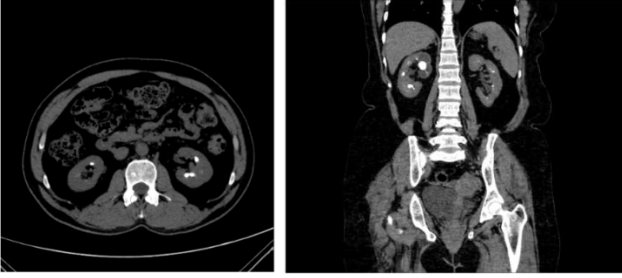


Fig. 3. Sample CT Images of Stone

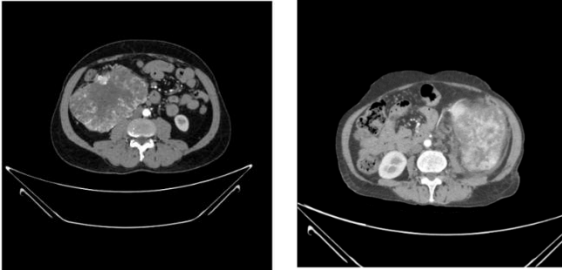


Fig. 4. Sample CT Images of Tumor

IV. METHODOLOGY

A. Data Preprocessing

Images were standardized to a resolution of 224X224 pixels. Given that CT scans do not rely on color features for diagnosis, the images were imported in grayscale to significantly reduce the model's parameter count. A rescaling layer normalized pixel values to a $[0, 1]$ range. The dataset was split into an 80% training and 20% validation set. To address the inherent class imbalance identified in Section III, balanced class weights were computed and applied during the training phase.

B. Proposed Custom CNN Architecture

We designed a customized, lightweight CNN optimized for grayscale medical image classification. The architecture features:

- **Feature Extraction:** Three sequential convolutional blocks containing Conv2D layers (32, 64, and 128 filters). Each block utilizes BatchNormalization for training stability, ReLU activation, and MaxPooling2D to downsample spatial dimensions.
- **Dimensionality Reduction:** A GlobalAveragePooling2D layer replaces the

traditional heavy dense/flatten layers, drastically reducing parameters and the risk of overfitting.

- **Classification Head:** A Dense layer of 256 units, followed by a Dropout rate of 0.3, and a final 4-unit Softmax layer for multi-class prediction.
- **Optimization Strategy:** The model was compiled using the Adam optimizer (learning rate = $1e-4$) and Categorical Crossentropy loss. EarlyStopping (patience=5) and ReduceLROnPlateau (factor=0.5, patience=3) were utilized to optimize convergence.

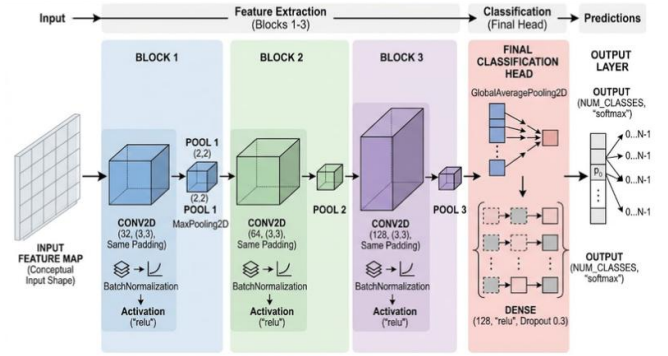


Fig. 5. Proposed Custom CNN Architecture

C. Baseline Architecture

To establish a performance baseline, three widely recognized architectures were trained on the dataset:

a) **VGG16:** A deep architecture utilizing 13 convolutional layers with fixed 3x3 filters.

- **Classification Head:** After the final max-pooling layer, we integrated a Flatten layer followed by two Dense layers (4096 units each) with ReLU activation and a final 4-unit Softmax layer.

b) **ResNet-50:** A 50-layer deep residual network that introduces "skip connections" to allow gradients to flow through very deep layers without vanishing.

- **Classification Head:** We replaced the global average pooling and fully connected layers with a custom head: GlobalAveragePooling2D → Dense (1024) → Dropout (0.5) → Softmax (4).

c) **EfficientNet:** Unlike VGG or ResNet, EfficientNet uses "Compound Scaling" to balance depth, width, and resolution simultaneously, making it highly resource-efficient.

- **Classification Head:** To maintain the model's efficiency, we utilized a GlobalAveragePooling2D layer followed directly by a Dropout layer (0.2) and a 4-unit Softmax output.

V. EXPERIMENTAL RESULTS

A. Baseline Model Performance

The three baseline models—EfficientNetB0, ResNet-50, and VGG16—were evaluated to establish a performance benchmark. As shown in Table II, the models exhibited significant variance in computational efficiency and diagnostic reliability.

Model	Overall Accuracy	Precision	Recall	F1-score
VGG16	89.10%	88.48%	87.19%	86.63%
ResNet-50	69.76%	65.43%	67.17%	64.93%
EfficientNet	76.84%	72.56%	74.32%	72.39%

EfficientNetB0: While demonstrating the fastest training time due to its compound scaling efficiency, it struggled significantly with the 'Stone' class (Precision: 0.41), frequently confusing stones with cysts (240 instances).

ResNet-50: Performed poorly on this dataset, particularly in the 'Tumor' class (Accuracy: 43.86%). The high training time (5032.98s) did not translate to high accuracy, suggesting the architecture may be too complex for grayscale CT features without extensive fine-tuning.

VGG16: Emerged as the strongest baseline with 89.10% accuracy. However, it showed a critical weakness in 'Tumor' detection (Recall: 0.61), misclassifying 154 tumor images as cysts.

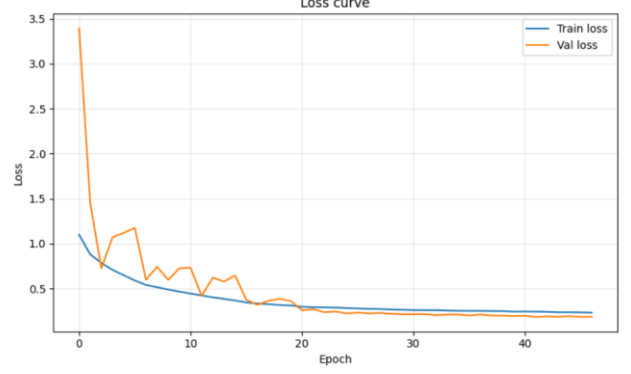
B. Proposed Custom CNN Performance

The proposed Custom CNN architecture significantly outperformed all baselines across every primary metric. By optimizing the filters for grayscale medical imaging and utilizing Global Average Pooling, the model achieved a superior diagnostic profile.

Class	Precision	Recall	F1-Score
Cyst	0.97	0.95	0.96
Normal	0.98	0.96	0.97
Stone	0.87	0.98	0.92
Tumor	0.93	0.94	0.93

- **Accuracy**: The proposed model reached 95.50% accuracy, a +6.4% improvement over the best baseline (VGG16).
- **Reliability**: Unlike the baselines, the custom model maintained balanced F1-scores (>0.92) across all four categories, effectively solving the Cyst/Tumor confusion that plagued the VGG and ResNet architectures.

C. Analysis of Training and Convergence



The training history reveals that the custom model reached convergence significantly faster than the baseline models. The inclusion of EarlyStopping and ReduceLROnPlateau successfully prevented overfitting, as evidenced by the validation loss closely tracking the training loss. The "Stone" class, which proved difficult for EfficientNet (0.41 precision), was handled with 0.87 precision in the custom model, likely due to the effectiveness of the BatchNormalization layers in stabilizing the feature maps for small calcification structures.

VI. CONCLUSION

This study highlights the effectiveness of lightweight, custom-built Convolutional Neural Networks for medical image classification over standard, heavy pre-trained models. Evaluated on a rigorously verified dataset of 12,446 CT images, our proposed architecture achieved an overall accuracy of 95.50%. Crucially, the model's extremely low inference time of 5.45 seconds demonstrates its viability for real-world, real-time clinical deployment, offering a highly efficient tool to assist medical professionals in the early detection of kidney abnormalities.

VII. REFERENCES

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